Atmospherically Deposited PBDEs, Pesticides, PCBs, and PAHs in Western US National Park Fish: Concentrations and Consumption Guidelines

Concentrations of polybrominated diphenyl ethers (PBDEs), pesticides, polychlorinated biphenyls (PCBs), and polycyclic aromatic hydrocarbons were measured in 136 fish from 14 remote lakes in 8 western US National Parks/Preserves between 2003 and 2005 and compared to human and wildlife contaminant health thresholds. A sensitive (median detection limit ‒18 pg/g wet weight), efficient (61% recovery at 8 ng/g), reproducible (4.1 %RSD), and accurate (7 % deviation from SRM) analytical method was developed and validated for these analyses. Concentrations of PCBs, hexachlorobenzene, hexachlorocyclohexanes, DDTs and chlordanes in western US fish were comparable to or lower than mountain fish recently collected from Europe, Canada, and Asia. Dieldrin and PBDE concentrations were higher than recent measurements in mountain fish and Pacific Ocean salmon. Concentrations of most contaminants in western US fish were 1–6 orders of magnitude below calculated recreational fishing contaminant health thresholds. However, contaminant concentrations exceeded subsistence fishing cancer screening values in 8 of 14 lakes. Average contaminant concentrations in fish exceeded wildlife contaminant health thresholds for piscivorous mammals in 5 lakes, and piscivorous birds in all 14 lakes. These results indicate that atmospherically deposited organic contaminants can accumulate in high elevation fish, reaching concentrations relevant to human and wildlife health.